

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to drive smarter business decisions.



What We Set Out to Understand



This project analyzes transactional data to reveal actionable insights across four key dimensions:

→ **Spending Patterns**

How much customers spend and what drives purchase amounts.

→ **Customer Segments**

Classifying buyers as New, Returning, or Loyal.

→ **Product Preferences**

Top-performing items by category and rating.

→ **Subscription Behavior**

How subscription status correlates with spending.

DATASET

Data at a Glance

3,900

Transactions

Across all product categories

18

Features

Demographics, purchases, and behavior

\$59.76

Avg. Purchase

Average spend per transaction

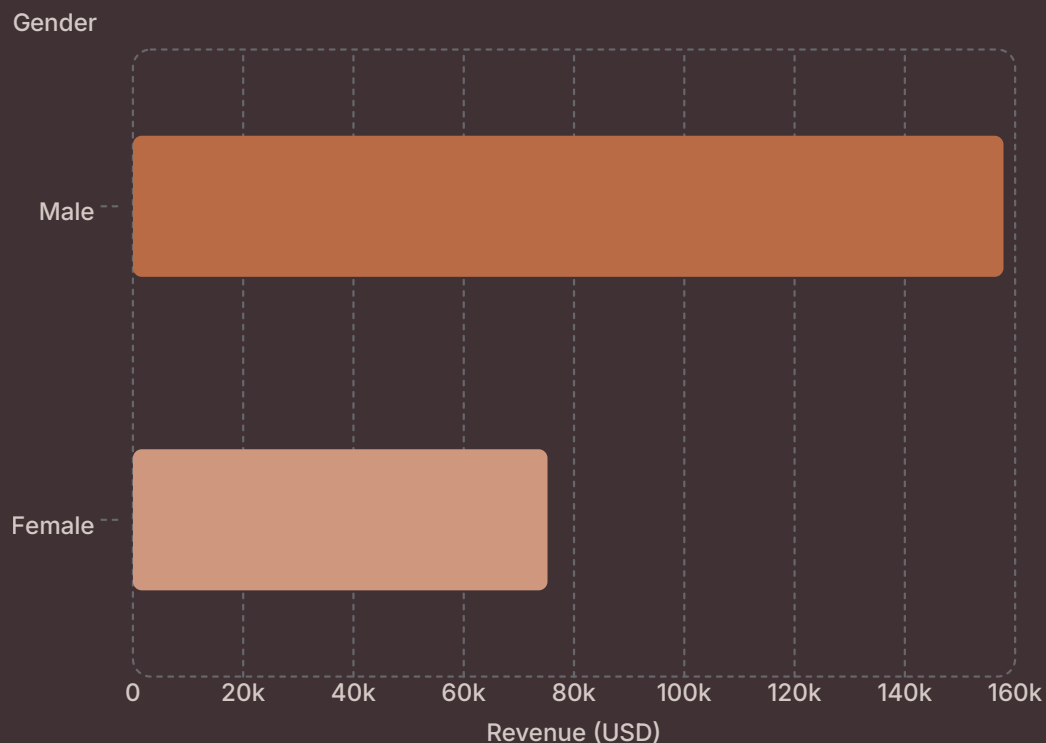
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Missing Values

In Review Rating, imputed by category median

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and behavioral signals (Discount Applied, Frequency, Review Rating, Shipping Type). Data was cleaned in Python, standardized to snake_case, and loaded into PostgreSQL for analysis.

Male Customers Drive 68% of Revenue



Key Finding

Male customers generated \$157,890 in total revenue — more than double the \$75,191 from female customers. Despite females representing a larger share of the customer base, male shoppers spend significantly more per transaction on average.

- ❏ This gap suggests a major opportunity to increase female customer engagement through targeted promotions or personalized product recommendations.

Top Products by Category & Rating

Top 5 Products by Average Rating

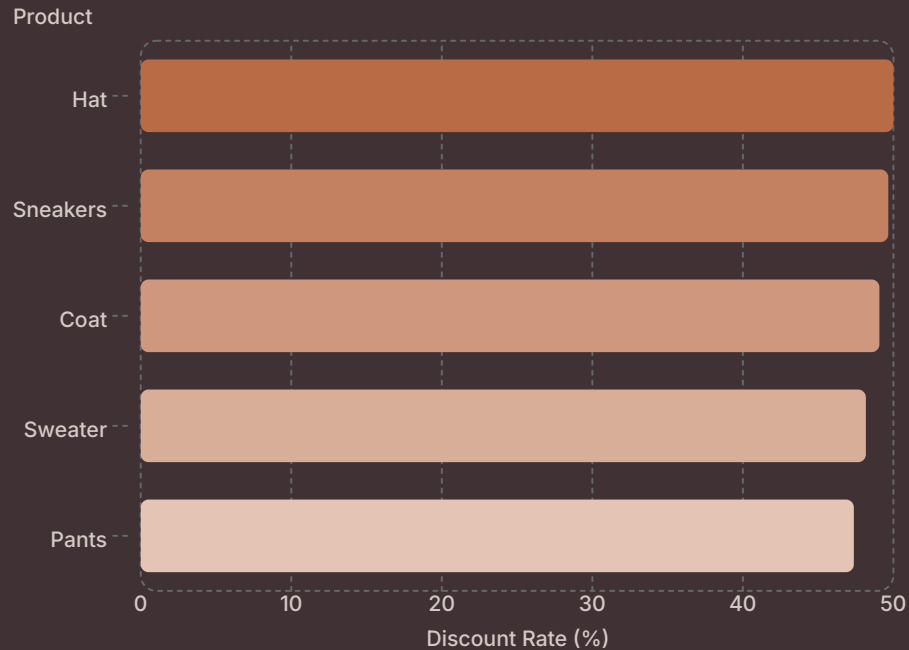
Item	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Purchased Items by Category

Category	Top Item	Orders
Clothing	Blouse	171
Accessories	Jewelry	171
Outerwear	Jacket	163
Footwear	Sandals	160

Gloves and Sandals lead in customer satisfaction, while Blouse and Jewelry dominate order volume — strong candidates for featured placements in marketing campaigns.

Discounts Are Concentrated in Outerwear & Accessories

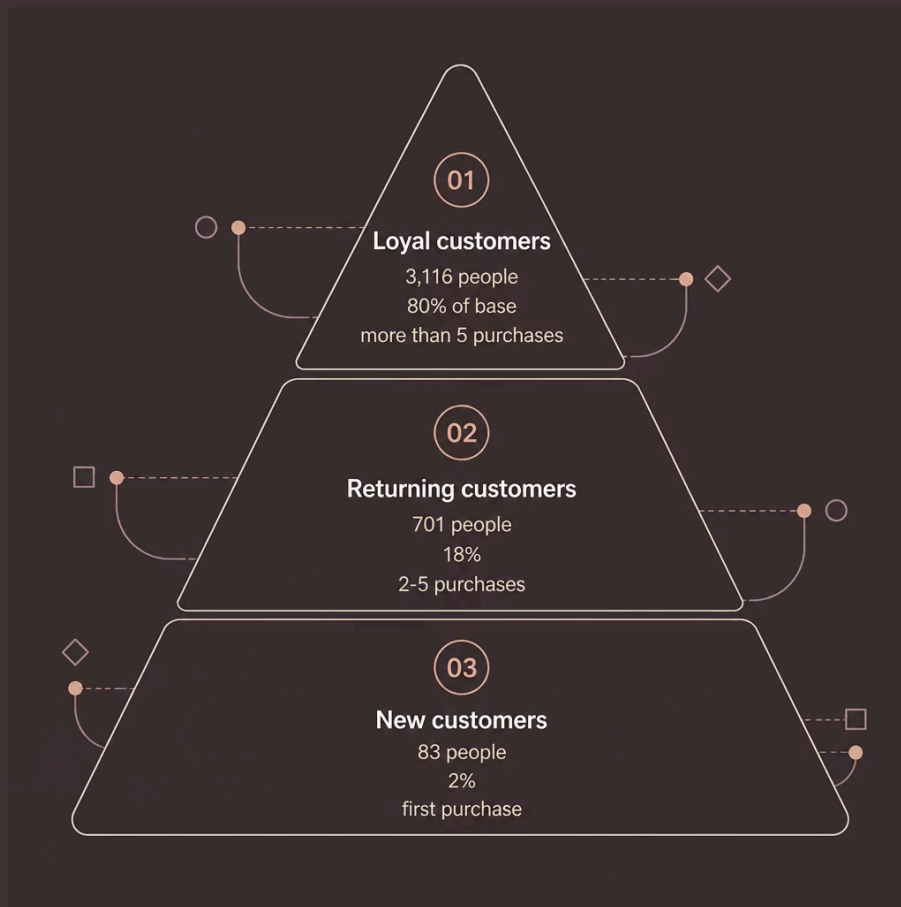


What This Reveals

The top 5 discount-dependent products all carry discount rates near 50%, suggesting these categories rely heavily on promotions to drive sales.

- **Hats** lead at exactly 50% — every other purchase is discounted
- Outerwear (Coat, Sweater) and Footwear (Sneakers) are similarly promotion-sensitive
- This pattern may erode margins over time if not carefully managed

Customer Segmentation: A Loyal-Heavy Base



Segment Breakdown

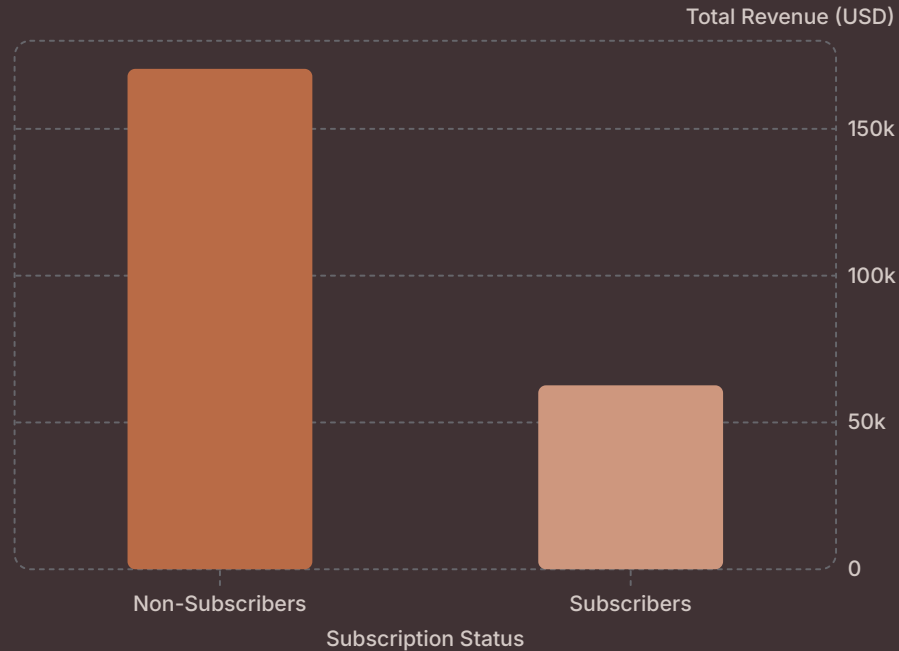
The customer base is overwhelmingly **Loyal** — 3,116 customers (80%) have made more than 5 purchases. Only 83 customers (2%) are classified as New.

Loyal — 3,116
80% of base

Returning — 701
18% of base

New — 83
2% of base

Subscribers Spend Slightly Less — But Represent Growth Potential



Subscription Insights

Only 27% of customers are subscribers, yet they represent a high-value retention opportunity. Non-subscribers generate \$170,436 in revenue vs. \$62,645 from subscribers.

- Average spend is nearly identical: \$59.87 (non-subscribers) vs. \$59.49 (subscribers)
- Of 958 repeat buyers (>5 purchases), only a portion have converted to subscriptions
- Converting loyal non-subscribers is the clearest revenue growth lever

Interactive Dashboard: Key Metrics at a Glance



Dashboard Highlights

The Power BI dashboard consolidates all SQL findings into an interactive, filterable view for stakeholders.

Clothing Dominates

\$100K revenue — 4×
Outerwear

Young Adults Lead

Highest revenue & sales by age
group

73% Non-Subscribers

Largest untapped conversion
opportunity

RECOMMENDATIONS

Five Actions to Drive Growth

1

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base, especially loyal repeat buyers.

2

Reward Loyalty

Launch a loyalty program to formally recognize and incentivize the 3,116 loyal customers.

3

Review Discount Policy

Audit the ~50% discount rates on Hats, Sneakers, and Outerwear to protect margins.

4

Amplify Top Products

Feature Gloves, Sandals, Blouse, and Jewelry prominently in campaigns based on ratings and volume.

5

Target High-Value Segments

Focus marketing on Young Adults, male shoppers, and express-shipping users for maximum ROI.